

JEE MAIN REVISION MASTER: REDOX REACTIONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 21 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Redox Reactions — Core Principles & Governing Laws	11 PYQs	Core Principle
High	Redox Reactions — Applied Cases, Reactions & Traps	11 PYQs	Applied Topics
Standard	Redox Reactions — Standard Calculations & Problem Archetypes	1 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Redox Reactions — Core Principles & Governing Laws	$2\text{Cu}^{2+} + 4\text{I}^{-} \longrightarrow 2\text{CuI}_{(s)}, \quad 2\text{Cl}^{-} \longrightarrow \text{Cl}_{(g)} + 2\text{e}^{-},$ $\begin{aligned} x-6 &= -1 \\ x &= -1+6=+5 \end{aligned}$
Redox Reactions — Applied Cases, Reactions & Traps	$\text{Cl}_{(g)} + \text{H}_{(g)}\text{O} \longrightarrow \text{HCl} + \text{HOCl}, \quad \text{KMnO}_{(4)},$ $2\text{MnO}_{(4)}^{-} + 3\text{SO}_{(2)} + 4\text{H}^{+} \longrightarrow 2\text{Mn}^{2+} + 3\text{SO}_{(2)}^{-} + 2\text{H}_{(2)}\text{O}$
Redox Reactions — Standard Calculations & Problem Archetypes	$\text{AgNO}_{(3)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Redox Reactions — Core Principles & Governing Laws (High Yield — 11 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Redox Reactions.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Redox Reactions.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Redox Reactions PYQs.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "In the titration of potassium permanganate ($\text{KMnO}_{(4)}$) against Ferrous ammonium sulphate (FAS) solution, dilute sulphuric acid but not HCl ..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Match List-I with List-II List-I (Types of redox reactions) List-II (Examples) a. Combination reaction i. $\text{Cl}_{(g)} + 2\text{Br}_{(aq)} \longrightarrow \dots$ " [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The reducing agent in the given equations $4\text{Ag}(s) + 8\text{CN}^{-}(aq) + 2\text{H}_{(2)}\text{O}(aq) + \text{O}_{(2)}(g) \rightarrow 4[\text{Ag}(\text{CN})_{(2)}]^{-}(aq) + 4\text{OH}^{-}(aq) + 2\text{Ag}(\dots)$ " [Ans: N/A]

3.2 Redox Reactions — Applied Cases, Reactions & Traps (High Yield — 11 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Redox Reactions.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $aC_2O_4^{2-} + bMnO_4^- + cH^+ \rightarrow xMn^{2+} + yH_2O + zCO_2$ a and ..." [Ans: A]
- [2026 Shift] Q#2 (MCQ): "Given below are two statements. Statement-I: In H_2O_2 , each oxygen atom is assigned an oxidation number of -1, in H_2O ..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Match List-I with List-II List-I (Types of redox reactions) List-II (Examples) a. Combination reaction i. $Cl_2(g) + 2Br_{(aq)}^- \rightarrow \dots$ " [Ans: N/A]

3.3 Redox Reactions — Standard Calculations & Problem Archetypes (Standard Yield — 1 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Redox Reactions.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2020 Shift] Q#2 (MCQ): "In which of the following cases a chemical reaction is possible?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$aC_2O_4^{2-} + bMnO_4^- + cH^+ \rightarrow xMn^{2+} + yH_2O + zCO_2$ a and x respectively are

- (A) 5,2 [Correct]
- (B) 4,1
- (C) 3,2
- (D) 4,2

Correct Answer: A

Solution Analysis: In acidic medium, MnO_4^- oxidises oxalate ion $C_2O_4^{2-}$ to CO_2 , and itself gets reduced to Mn^{2+} . First write the two half-reactions. For oxalate ion: $C_2O_4^{2-} \rightarrow 2CO_2$ Carbon is oxidised here, and the balanced half-reaction is: $C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$ For permanganate ion in acidic medium: $MnO_4^- \rightarrow Mn^{2+}$ Balancing oxygen and hydrogen using H_2O and H^+ : $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

Benchmark Problem 2: KCET 2025 (Q#3)

Match List-I with List-II List-I (Types of redox reactions) List-II (Examples) a. Combination reaction i. $\text{Cl}_2(\text{g}) + 2\text{Br}^-(\text{aq}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{Br}_2(\text{l})$ b. Decomposition reaction ii. $2\text{H}_2\text{O}(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ c. Displacement reaction iii. $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ d. Disproportionation reaction iv. $2\text{H}_2\text{O}_2(\text{l}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ Choose the correct answer from the options given below.

- (A) a-iv, b-iii, c-i, d-ii
- (B) a-ii, b-i, c-iv, d-iii
- (C) a-iii, b-iv, c-i, d-ii
- (D) a-iii, b-ii, c-i, d-iv

Correct Answer: None

Solution Analysis: Combination reaction - no example given Decomposition reaction - $\text{H}_2\text{O}(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ Displacement reaction - $\text{Cl}_2(\text{g}) + 2\text{Br}^-(\text{aq}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{Br}_2(\text{l})$ Disproportionation reaction - $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$

Benchmark Problem 3: KCET 2020 (Q#2)

In which of the following cases a chemical reaction is possible?

- (A) $\text{ZnSO}_4(\text{aq})$ is placed in a copper vessel
- (B) AgNO_3 solution is stirred with a copper spoon
- (C) Conc. HNO_3 is stored in a platinum vessel
- (D) gold ornaments are washed with dil. HCl

Correct Answer: None

Solution Analysis: A chemical reaction is possible when AgNO_3 solution is stirred with copper spoon as copper has lesser positive value of standard electrode reduction potential as compared to Ag. The reaction will be, $2\text{AgNO}_3 + \text{Cu} \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{Ag}$